

CodeAlpha Internship

Task 2: Exploratory Data Analysis (EDA)

Submitted By

- **Name:** Avinash Sudake
 - **Domain:** Data Analytics
 - **Internship:** CodeAlpha Data Analytics Internship
 - **Task:** Exploratory Data Analysis (EDA)
 - **Tools Used:** Python, Pandas, Matplotlib, Google Colab
 - **Dataset:** Supermarket Sales Dataset
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1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying advanced analysis techniques. The main purpose of EDA is to identify patterns, trends, relationships, and insights from the data using statistical methods and visualizations.

2. Objective

The objective of this project is to analyze supermarket sales data and identify useful business insights by using Python libraries such as Pandas and Matplotlib.

3. Dataset Information

Dataset Name: Supermarket Sales Dataset

Total Records: 1000

Total Columns: 17

File Format: CSV

The dataset contains information about customer purchases, product categories, payment methods, sales, tax, ratings, and branch details.

4. Tools and Libraries Used

- Google Colab

- Python
- Pandas
- Matplotlib

5. Data Cleaning

The dataset was checked for quality before analysis.

- No Missing Values found.
- No Duplicate Records found.
- Date column converted into Date format.
- Dataset was ready for analysis.

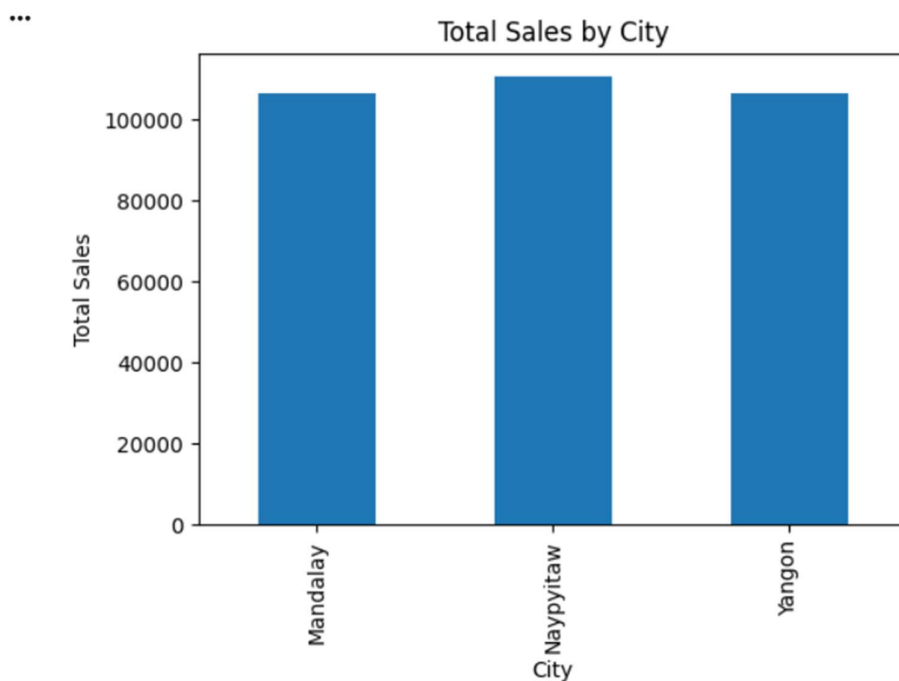
6. Exploratory Data Analysis

The following analyses were performed:

City-wise Sales Analysis

A bar chart was created to compare total sales across different cities.

Chart: Total Sales by City



Observation:

Naypyitaw recorded the highest total sales.

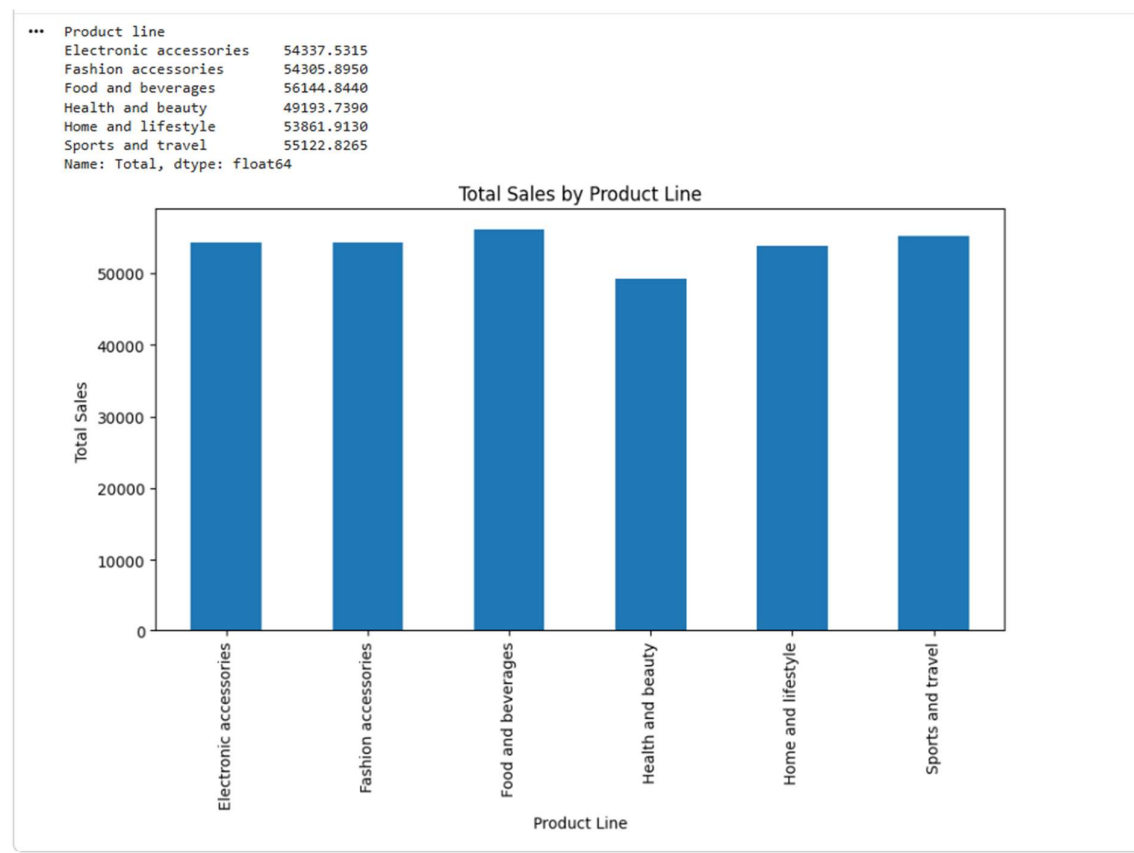
Insight:

This indicates that Naypyitaw performed better than the other cities in terms of revenue generation.

Product Line Analysis

A bar chart was used to compare sales across product categories.

Chart: Total Sales by Product Line



Observation:

Food and Beverages generated the highest sales.

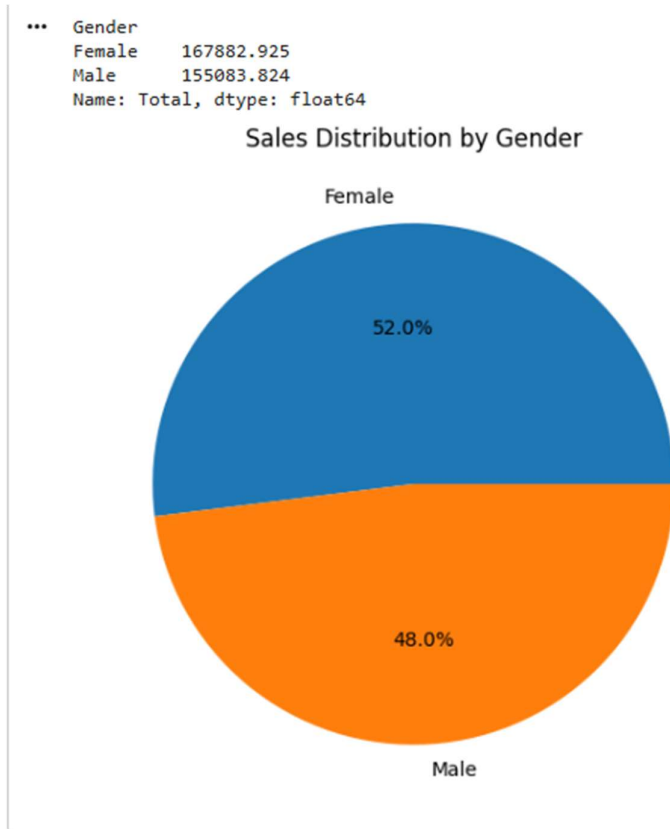
Insight:

Customers purchased Food and Beverages more frequently than other product categories.

Gender-wise Sales Analysis

A pie chart was created to compare sales by gender.

Chart: Sales Distribution by Gender



Observation:

Female customers contributed slightly more sales than male customers.

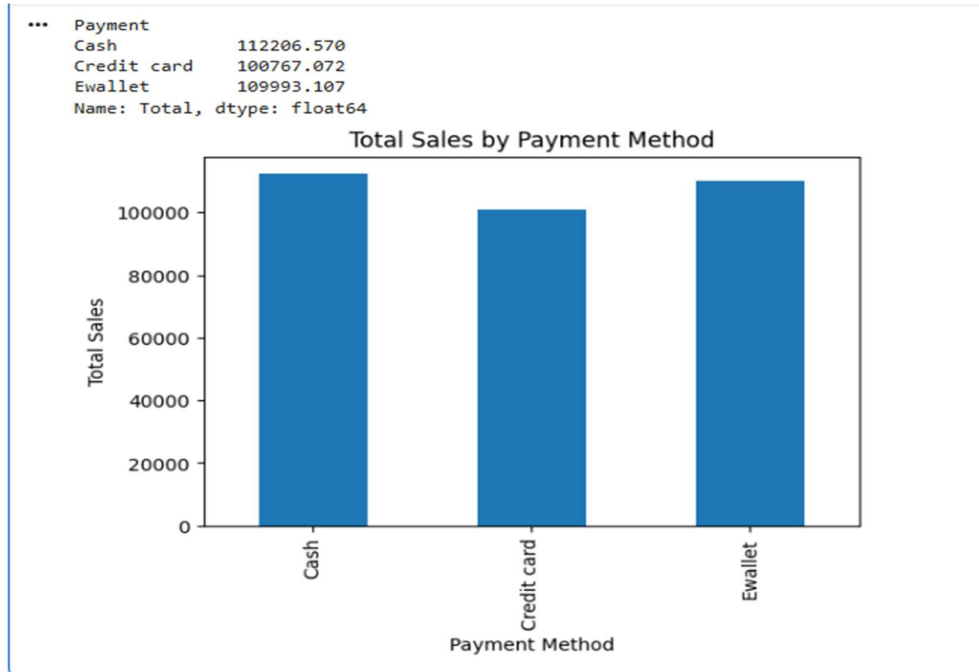
Insight:

Female customers generated 52% of total sales, showing a slightly higher purchasing contribution.

Payment Method Analysis

A bar chart was created to compare different payment methods.

Chart: Total Sales by Payment Method



Observation:

Cash was the most preferred payment method.

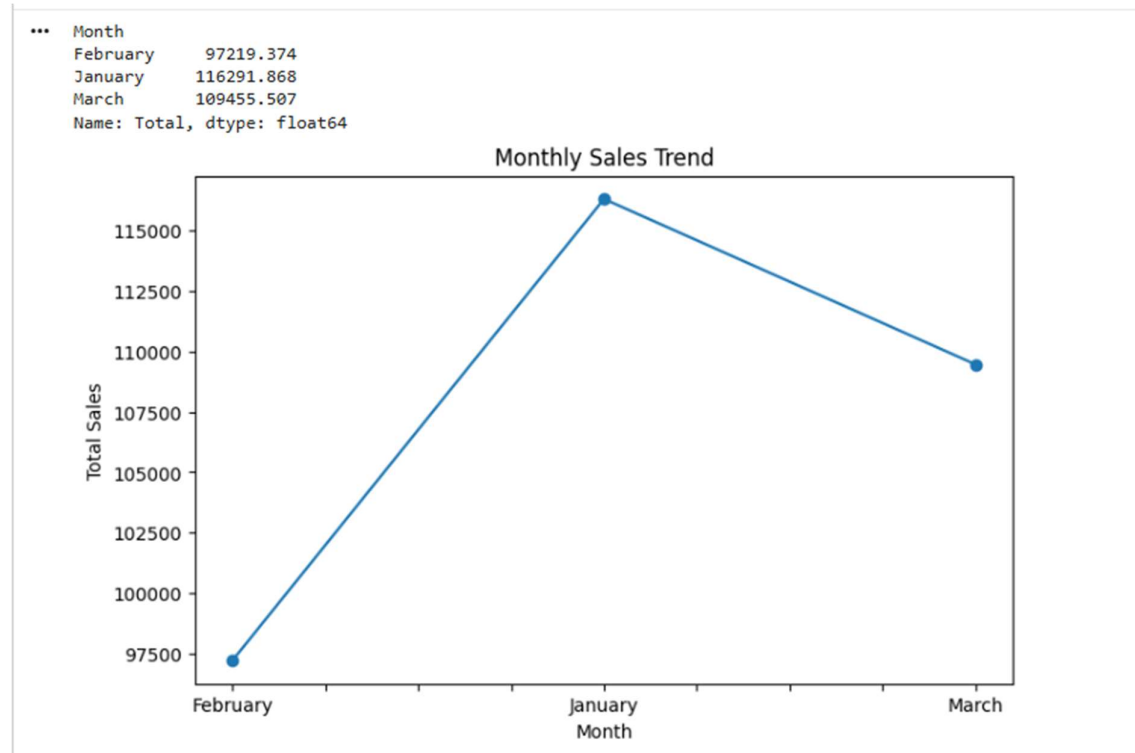
Insight:

Most customers preferred cash payments over Credit Card and E-wallet.

Monthly Sales Trend

A line chart was used to analyze monthly sales.

Chart: Monthly Sales Trend



Observation:

January recorded the highest sales.

Insight:

Sales decreased in February and increased again in March.

7. Final Insights

1. Naypyitaw recorded the highest sales.
2. Food and Beverages generated the highest revenue.
3. Health and Beauty generated the lowest sales.

4. Female customers contributed 52% of total sales.
 5. Cash was the most preferred payment method.
 6. January recorded the highest monthly sales.
 7. No missing values were found.
 8. No duplicate records were found.
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8. Conclusion

This Exploratory Data Analysis (EDA) project successfully analyzed the Supermarket Sales Dataset using Python, Pandas, and Matplotlib. The analysis identified important sales trends, customer behavior, and business patterns through statistical summaries and visualizations. The dataset contained no missing values or duplicate records, making it reliable for analysis. The insights obtained from this project can help businesses make better decisions, improve sales strategies, and enhance customer satisfaction.

9. References

- Supermarket Sales Dataset
 - Python Documentation
 - Pandas Documentation
 - Matplotlib Documentation
 - Google Colab
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